Computer Science 1

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2026-09-04

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Contact

- **Year**: 2026/27/1, Semester 1
- **Class Location**: P/106/A, Institute of Psychology, University of Debrecen
- **Meeting Times**: Tuesday 10:00 - 11:40
- **Contact Information**: Please contact me through email anytime: abari.kalman@arts.unideb.hu
- **Registration table for homework assignments and grades**: link to the spreadsheet

1 Syllabus

Course Number: BTPS1210BA-E
Credits: 2
Course type: Practical
Year: 2026/27/1, Semester 1 (Autumn)
Class Location: P/106/A, Institute of Psychology, University of Debrecen
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1.1 Course overview

During the course, students will acquire the basic digital skills they will need for their studies. The course attempts to balance the knowledge levels of students with different backgrounds. At the beginning of the course, students will learn basic methods of word processing, spreadsheet and presentation design.

The main purpose of the course is to support statistics courses. Today, statistical analysis is carried out by computer, so knowledge of one (or more) statistical software packages is essential. The rest of the course, students will be introduced to the *jamovi* statistical software and will also be introduced to the use of *R/RStudio*. The *jamovi* part of the course will cover statistical processes of data entry and data preparation, descriptive analysis, and graphic analysis. In *RStudio* students will be able to create markdown-based documents and generate HTML documents.

1.2 Attendance

Active participation is an important part of this class. Attendance are required. The maximum number of absences from the practical sessions is three. If the student exceeds the maximum number of absences, a practical grade will not be given and a "refused" will be entered in the electronic learning system.

1.3 Assessment and Grades

During the semester there will be 4 homework assignments and one project. In the Table 1.1 you can see the details of the homework assignments and the project. The project will be presented during the last two classes of the semester. The Table 1.1 also shows the deadlines for the homework assignments and the project.

Table 1.1: Points and deadlines

Components	Points	Links	Dates
Homework 1 • RStudio reporting: markdown • Quarto document	15	Details[1]	October 14.
Homework 2 • Jamovi (1) • JabRef	15	Details[2]	November 4.
Homework 3 • Jamovi (2)	15	Details[3]	November 25.
Homework 4 • Word • JabRef	15	Details[4]	December 7.
Project final artifact	30	Details[5]	November 9. and December 5.
Project final presentation	10	Details	Last two classes of the semester.
TOTAL	100		

Many ways to earn **extra points**, which are added to the points shown in the Table 1.1. Extra points can be earned in the following ways:

[1]Hint: My solution of Homework 1
[2]Hint: My solution of Homework 2
[3]Hint: My solution of Homework 3
[4]Hint: My solution of Homework 4
[5]Hint: My solution of Project final artifact. My solution of The answer.

- active participation in class (up to 5 points)
- making and presenting a Kahoot! test (3 points each)
- winning a Kahoot! test (3 points each)
- sending a solution of a lab
  - Lab 01
    * Problem 1: up to 1 point (link)
    * Problem 2: up to 1 point (link)
    * Problem 3: up to 2 points (link)
  - Lab 02
    * Problem 1: up to 2 points (link)
    * Problem 2: up to 2 points (link)
    * Problem 3: up to 3 points (link)
    * Problem 4: up to 3 points (link)
  - Lab 03
    * Problem 1: up to 2 points (link)
    * Problem 2: up to 2 points (link)
    * Problem 3: up to 3 points (link)
  - Lab 04
    * Problem 1: up to 2 points (link)
    * Problem 2: up to 2 points (link)
    * Problem 3: up to 3 points (link)
    * Problem 4: up to 3 points (link)
  - Lab 06
    * Problem 1: up to 2 points (link)
    * Problem 2: up to 2 points (link)
    * Problem 3: up to 3 points (link)
    * Problem 4: up to 3 points (link)
  - Lab 07
    * Problem 1: up to 2 points (link)
    * Problem 2: up to 2 points (link)
    * Problem 3: up to 3 points (link)
    * Problem 4: up to 3 points (link)
    * Problem 5: up to 3 points (link)
    * Problem 6: up to 3 points (link)

    - Problem 7: up to 3 points (link)
    - Problem 8: up to 3 points (link)
  - Lab 08
    - Problem 1: up to 2 points (link)
    - Problem 2: up to 2 points (link)
    - Problem 3: up to 3 points (link)
    - Problem 4: up to 3 points (link)
    - Problem 5: up to 3 points (link)
  - Lab 09
    - Problem 1: up to 2 points (link)
    - Problem 2: up to 2 points (link)
    - Problem 3: up to 3 points (link)
    - Problem 4: up to 3 points (link)
  - Lab 10
    - Problem 1: up to 4 points (link)
    - Problem 2: up to 4 points (link)
  - Lab 11
    - Problem 1: up to 4 points (link)
    - Problem 2: up to 4 points (link)

The Table 1.2 shows the grading scale. A student's final grade will be determined by the total number of points earned during the semester. The final grade will be determined as follows:

Table 1.2: Grading

Points	Grade offered
0–50	1 (failed)
51–60	2
61–74	3
75–85	4
86–	5

Students can earn a maximum of 100 points through the homework assignments and the project work, to which extra points are added.

1.4 Course details

Table 1.3: Course details

Day	Topic	Links
Sept 8	Intro	Details
Sept 15	Files, directories, compressing	Details
Sept 22	RStudio reporting: markdown, Quarto document	Details
Sept 29	JabRef - Bibliography management	Details
Oct 6	TXT, CSV, Spreadsheets	Details
Oct 13	Jamovi - Introduction	Details
Oct 20	Autumn break	
Oct 27	Jamovi - Import data, Data conversion	Details
Nov 3	Jamovi - Data transformation	Details
Nov 10	Jamovi - Descriptive statistics I.	Details
Nov 17	Jamovi - Descriptive statistics II.	Details
Nov 24	Word and PowerPoint	Word; PPT
Dec 1	Project presentation	Details
Dec 8	Project presentation	Details

1.5 Textbooks

- **Required:**
  - University of Helsinki (2023, p. 1 Introduction, Advanced studies: presenting data)
  - Navarro & Foxcroft (2022, pp. 1–117)
  - Poulson (2019, Getting started, Wrangling data, Exploration)
- **Optional texts:**
  - Pajankar (2022)
  - Cote et al. (2021, pp. 1–77)
  - Gordon & Bataineh (2022, pp. 1–109)
  - Landers (2024)

1.6 Software

The course requires some software.

1.7 Prerequisites

No prerequisite is required. Prior knowledge in computer literacy (word processing, spreadsheets, presentation) is recommended but not required.

2 Course details

2.1 Intro

- Readings:
  - Navarro & Foxcroft (2022, pp. 3–12)

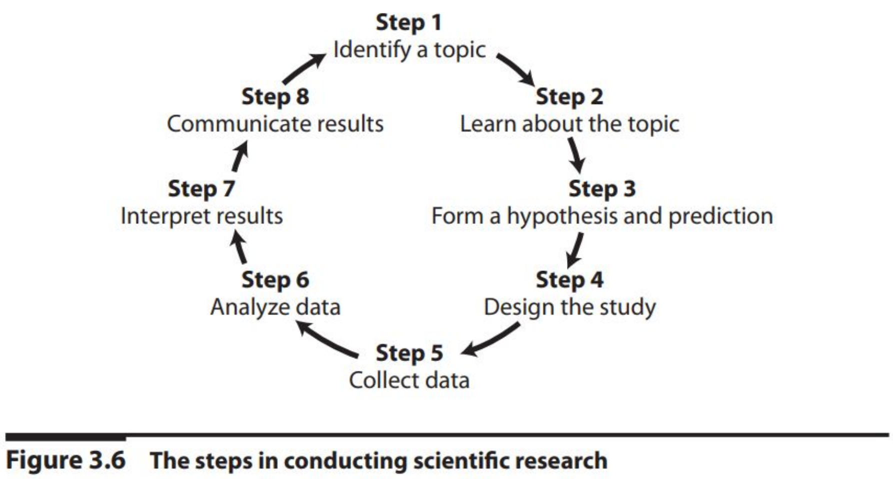

Figure 2.1: Schweigert (2021) pp. 58

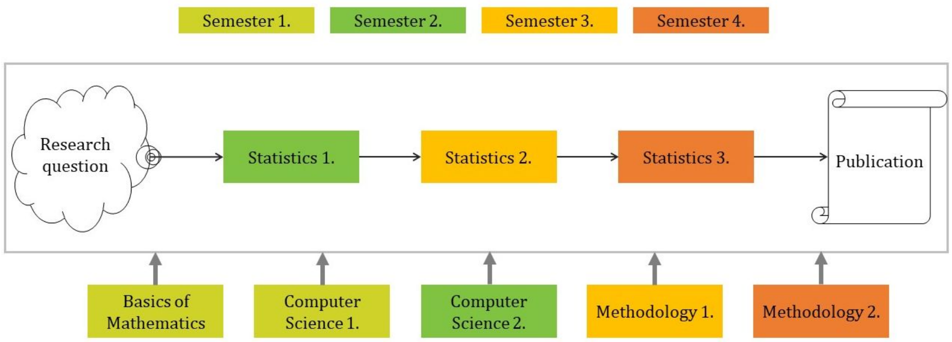

Figure 2.2: The course in the BA program

2.2 Files, directories, compressing

- Readings:
  - University of Helsinki (2023) 1.1 Computer functionality
  - University of Helsinki (2023) 1.2. Using files and programs
- Problems
  - Lab 01
- Questions
  - What is the purpose of buying a computer?
  - What problems does the computer solve?
  - How does a computer solve problems?
  - What is the file?
  - What is the purpose of the file?
  - What is the directory
  - What is the purpose of the directory?
  - What is the structure of directories?
  - What is packing and unpacking (or compressing and extraction)?
  - What is the purpose of the packing?
- Skills
  - creating, renaming, coping, moving, deleting files and directories
  - compressing and extracting files and directories
  - creating file with special type (TXT, DOCX, XLSX, ODT, ODS, PDF, PPTX, PNG, JPG, OMV, BIB, etc.)

    - typing special characters (ampersand, angle brackets, apostrophe, asterisk, at sign, backslash, braces, brackets, caret, dollar sign, em dash, en dash, hyphen, backtic, tilde, parentheses, percent, pipe, slash, underscore, etc.)

2.3 RStudio reporting

- Readings:
    - RStudio for the Total Beginner (only the first minute and a half)
    - Posit Cheatsheets: RStudio IDE
    - The Markdown Guide: Getting Started, Markdown Cheat Sheet, Basic Syntax, Extended Syntax
    - Posit Cheatsheets: Publish and Share with Quarto
    - Tutorial: Hello, Quarto
- Problems
    - Lab 02

2.4 JabRef - Bibliography management

- Readings:
    - JabRef: Getting started
    - Quarto: Citations & Footnotes
- Problems
    - Lab 03

2.5 TXT, CSV, Spreadsheets

- Readings:
    - What is Tabular Data?
    - University of Helsinki (2023) P.3 Spreadsheets
- Problems
    - Lab 04

2.6 Jamovi - Introduction

- Readings:
  - Navarro & Foxcroft (2022, pp. 43–49)
  - Poulson (2019)
    - Welcome
    - Installing jamovi
    - Navigating jamovi
- Problems
  - Lab 05

2.7 Jamovi - Import data, Data conversion

- Readings:
  - Navarro & Foxcroft (2022, pp. 49–54)
  - Poulson (2019)
    - Sample data
    - jamovi modules
    - Entering data
    - Importing data
    - Variable types & labels
- Problems
  - Lab 06

2.8 Jamovi - Data transformation

- Readings:
  - Navarro & Foxcroft (2022, pp. 97–117)
  - Poulson (2019)
    - Wrangling data: chapter overview
    - Computing means
    - Computing z-scores
    - Transforming scores to categories
    - Filtering cases
- Problems
  - Lab 07

2.9 Jamovi - Descriptive statistics I.

- Readings:
  - Navarro & Foxcroft (2022, pp. 59–84)
  - Poulson (2019)
    - Exploration: chapter overview
    - Descriptive statistics
- Problems
  - Lab 08

2.10 Jamovi - Descriptive statistics II.

- Readings:
  - Navarro & Foxcroft (2022, pp. 85–96)
  - Poulson (2019)
    - Histograms
    - Density plots
    - Box plots
    - Violin plots
    - Dot plots
    - Bar plots
    - Exporting tables & plots
- Problems
  - Lab 09

2.11 Word processing - Styles, TOC, Bibliography, APA7

- Readings:
  - University of Helsinki (2023) P.2 Word processing
  - Study skills: APA7
  - Citation: About APA 7th ed.
- Problems
  - Lab 10

2.12 Presentation: PowerPoint

- Readings:
  - University of Helsinki (2023) P.4 Slide shows
- Problems
  - Lab 11

2.13 Project presentation

3 Software

Please install the following software on your computer:

- **Jamovi** [Available from url: https://www.jamovi.org/download.html. Choose the *current* version.]
- **R** [Available from url: https://cloud.r-project.org/.]
- **RStudio** [Available from url: https://posit.co/downloads/. Choose the free *RStudio Desktop*.]
- **JabRef** [Available from url: https://www.jabref.org/#download.]
- **Zotero** [Available from url: https://www.zotero.org/downloads. *Zotero* is an alternative to *JabRef*.]
- **Mendeley Desktop** [Available from url: https://www.mendeley.com/download-reference-manager. *Mendeley Desktop* is an alternative to the *JabRef*. We won't use it in the course, but it's a good reference manager.]

4 Schedule of the Academic Year

The academic year in Hungary starts in September and is built up of two semesters: autumn and spring semesters. Both semesters start with a registration period which is followed by the study period and an examination period afterwards. Details in Hungarian.

I. (AUTUMN) SEMESTER

- Free course registration and withdrawal period: 31 August, 2026 (Monday), 8:00 – 20 September, 2026 (Sunday), 23:59 (3 weeks)
  - Non-graduating students may register for courses from 13:00 on 1 September, 2026.
- Registration week: 31 August – 4 September, 2026 (1 week)
- Part-time education begins on 3 September, 2026 (Thursday).
- Study period:
  - Non-graduating students: 7 September – 11 December, 2026 (13 weeks)
  - Graduating students completing their studies in January 2027: 7 September – 27 November, 2026 (11 weeks)
- Autumn break: 19 – 22 October, 2026 (1 week)
- Examination period:
  - Non-graduating students: 14 December, 2026 – 29 January, 2027 (7 weeks)
  - Final examination period for graduating students completing their studies in January 2027: 28 December, 2026 – 15 January, 2027 (3 weeks)

II. (SPRING) SEMESTER

- Registration week: 1 – 5 February, 2027 (1 week)
- Study period for non-graduating students: 8 February – 14 May, 2027 (13 weeks)
- Spring break: 30 March – 2 April, 2027 (1 week)
- Examination period for non-graduating students: 18 May – 2 July, 2027 (7 weeks)

References

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Poulson, B. (2019). *Jamovi.* https://datalab.cc/jamovi/

Schweigert, W. A. S. (2021). *Research methods in psychology: A handbook* (Fourth Edition). Waveland Press.

University of Helsinki. (2023). *Student's digital skills.* https://blogs.helsinki.fi/students-digital-skills/